

Homework 3

1

A synthetic fiber used in manufacturing carpet has tensile strength that is normally distributed with mean 75.5 psi and standard deviation 3.5 psi. Find the probability that a random sample of $n = 6$ fiber specimens will have a sample mean tensile strength that exceeds 75.75 psi.

Hint: Even though the sample size is only 6 you may assume that it is large enough to justify the use of the Central Limit Theorem.

✓ Answer ✓

$$X \sim N(75.5, 3.5^2)$$

$$\bar{X} \sim N\left(75.5, \frac{3.5^2}{6}\right) \text{ by the CLT for an average of 6 samples}$$

We are finding $P(\bar{X} > 75.75)$.

$$P(\bar{X} > 75.75) = P\left(Z > \frac{75.75 - 75.5}{\sqrt{\frac{3.5^2}{6}}}\right) = P(Z > 0.1749635530559413)$$

$$= P(Z < -0.1749635530559413) \text{ by } Z \text{ symmetry}$$

$$= 0.43055413611371507 \text{ by normal CDF function}$$

2

Suppose we have a random sample of size $2n$ from a population denoted by X , and $E(X) = \mu$ and $V(X) = \sigma^2$. Let

$$\bar{X}_1 = \frac{1}{2n} \sum_{i=1}^{2n} X_i$$

$$\bar{X}_2 = \frac{1}{n} \sum_{i=1}^n X_i$$

be two estimators of μ . Which is the better estimator of μ ? Explain your choice.

✓ Answer

$$E(\bar{X}_1) = E\left(\frac{1}{2n} \sum_{i=1}^{2n} X_i\right) = \frac{1}{2n} \sum_{i=1}^{2n} E(X_i) \text{ assuming samples are independent}$$

$$\frac{1}{2n} \sum_{i=1}^{2n} \mu = \frac{1}{2n} 2n\mu = \mu$$

$$E(\bar{X}_2) = E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n} \sum_{i=1}^n E(X_i) \text{ assuming samples are independent}$$

$$\frac{1}{n} \sum_{i=1}^n \mu = \frac{1}{n} n\mu = \mu$$

$Var(\bar{X}_1) = Var\left(\frac{1}{2n} \sum_{i=1}^{2n} X_i\right) = \left(\frac{1}{2n}\right)^2 \sum_{i=1}^{2n} Var(X_i)$ assuming samples are independent

$$\left(\frac{1}{2n}\right)^2 \sum_{i=1}^{2n} \sigma^2 = \left(\frac{1}{2n}\right) 2n\sigma^2 = \frac{\sigma^2}{2n}$$

$Var(\bar{X}_2) = Var\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \left(\frac{1}{n}\right)^2 \sum_{i=1}^n Var(X_i)$ assuming samples are independent

$$\left(\frac{1}{n}\right)^2 \sum_{i=1}^n \sigma^2 = \left(\frac{1}{n}\right) n\sigma^2 = \frac{\sigma^2}{n}$$

I would choose $\bar{X}_1$ as both means are the same and unbiased, and $\bar{X}_1$ has lower variance.

3

Let X and Y be bivariate normal random variables with:

$$\mu_X = 2$$

$$\mu_Y = 3$$

$$\sigma_X^2 = 1.44$$

$$\sigma_Y^2 = 4.84$$

$$\rho = 0.6$$

$$W = 0.5X + 2Y$$

Find:

a

The distribution of W

✓ Answer

Since a linear combination of normally distributed random variables is also normally distributed, W is normally distributed.

$$\begin{aligned} E(W) &= E(0.5X + 2Y) = 0.5E(X) + 2E(Y) \\ &= 0.5\mu_X + 2\mu_Y = 0.5(2) + 2(3) = 1 + 6 = 7 \end{aligned}$$

$$Cov(X, Y) = \rho\sigma_X\sigma_Y$$

$$\begin{aligned} Var(W) &= Var(0.5X + 2Y) = 0.5^2 Var(X) + 2^2 Var(Y) + 2(0.5)(2)Cov(X, Y) \\ &= 0.5^2(1.44) + 2^2(4.84) + 2(0.5)(2)\rho\sigma_X\sigma_Y \\ &= 0.5^2(1.44) + 2^2(4.84) + 2(0.6)\sqrt{(1.44)(4.84)} \\ &= 22.888 \end{aligned}$$

$$\therefore W \sim N(7, 22.888)$$

b

$$P(W > 7)$$

$$\begin{aligned}
 P(W > 7) &= P(Z > \frac{7-7}{\sqrt{22.888}}) = P(Z > 0) \text{ by } Z \text{ transform} \\
 &= P(Z < 0) \text{ by } Z \text{ symmetry} \\
 &= 0.5 \text{ by } Z \text{ table}
 \end{aligned}$$

4

Let X_1, X_2, X_3 be independent random variables such that:

$$\begin{aligned}
 E(X_1) &= 1 \\
 E(X_2) &= -3 \\
 E(X_3) &= 5 \\
 Var(X_1) &= 4 \\
 Var(X_2) &= 1 \\
 Var(X_3) &= 6.25
 \end{aligned}$$

Find:

a

$$E(3X_1 + 2X_2 - 0.5X_3)$$

✓ **Answer**

$$\begin{aligned}
 &= 3E(X_1) + 2E(X_2) - 0.5E(X_3) \\
 &= 3(1) + 2(-3) - 0.5(5) \\
 &= -5.5
 \end{aligned}$$

b

$$Var(3X_1 + 2X_2 - 0.5X_3)$$

✓ **Answer**

$$\begin{aligned}
 &= 3^2 Var(X_1) + 2^2 Var(X_2) + (-0.5)^2 Var(X_3) \\
 &= 3^2(4) + 2^2(1) + (-0.5)^2(6.25) \\
 &= 41.5625
 \end{aligned}$$

5

Suppose X_1 and X_2 are independent with the same but unknown mean μ and variances, with point estimators:

$$\begin{aligned}
 Var(X_1) &= 15 \\
 Var(X_2) &= 4 \\
 \hat{\mu}_1 &= \frac{X_1}{2} + \frac{X_2}{2} \\
 \hat{\mu}_2 &= \frac{X_1}{4} + \frac{3X_2}{4} \\
 \hat{\mu}_3 &= \frac{X_1}{6} + \frac{X_2}{3} + 9
 \end{aligned}$$

a

Are any of the estimators unbiased?

✓ **Answer**

$$E(\hat{\mu}_1) = E\left(\frac{X_1}{2} + \frac{X_2}{2}\right) = 0.5E(X_1) + 0.5E(X_2) = 0.5(\mu) + 0.5(\mu) = \mu$$

$$E(\hat{\mu}_2) = E\left(\frac{X_1}{4} + \frac{3X_2}{2}\right) = 0.25E(X_1) + 0.75E(X_2) = 0.25\mu + 0.75\mu = \mu$$

$$E(\hat{\mu}_3) = E\left(\frac{X_1}{6} + \frac{X_2}{3} + 9\right) = \frac{E(X_1)}{6} + \frac{E(X_2)}{3} + 9 = \frac{\mu}{6} + \frac{\mu}{3} + 9 = \frac{\mu}{2} + 9$$

Yes, $\hat{\mu}_3$ is biased as its expectation is not equal to μ .

b

Which one has the smallest variance?

✓ **Answer**

$$Var(\hat{\mu}_1) = Var\left(\frac{X_1}{2} + \frac{X_2}{2}\right) = 0.5^2 E(X_1) + 0.5^2 E(X_2) = 0.5^2(15) + 0.5^2(4) = \frac{19}{4}$$

$$Var(\hat{\mu}_2) = Var\left(\frac{X_1}{4} + \frac{3X_2}{2}\right) = 0.25^2 E(X_1) + 0.75^2 E(X_2) = 0.25^2(15) + 0.75^2(4) = 3.1875$$

$$Var(\hat{\mu}_3) = Var\left(\frac{X_1}{6} + \frac{X_2}{3} + 9\right) = \frac{E(X_1)}{6^2} + \frac{E(X_2)}{3^2} = \frac{15}{6^2} + \frac{4}{3^2} = 0.8611111111111112$$

$\hat{\mu}_3$ has the smallest variance, but it is biased, $\hat{\mu}_2$ has the smallest unbiased variance.

c

Calculate the mean square error (MSE) of the estimators.

✓ **Answer**

$$MSE = Var + Bias^2$$

$$MSE(\hat{\mu}_1) = \frac{19}{4} + 0^2 = \frac{19}{4}$$

$$MSE(\hat{\mu}_2) = 3.1875 + 0^2 = 3.1875$$

$$MSE(\hat{\mu}_3) = 0.8611111111111112 + \left(9 - \frac{\mu}{2}\right)^2$$

Depending on μ , $\hat{\mu}_2$ or $\hat{\mu}_3$ could be lower (better).

6

Let $X_1, X_2, \dots, X_7$ denote a random sample from a population having mean μ and variance σ^2 .

Consider the following estimators of μ :

$$\hat{\Theta}_1 = \frac{X_1 + X_2 + \dots + X_7}{7}$$

$$\hat{\Theta}_2 = \frac{2X_1 - X_6 + X_4}{2}$$

a

Is either estimator unbiased? Justify your answer.

✓ **Answer**

$$\begin{aligned} E(\hat{\Theta}_1) &= E\left(\frac{X_1 + X_2 + \dots + X_7}{7}\right) = \frac{1}{7}(E(X_1) + E(X_2) + \dots + E(X_7)) \\ &= \frac{1}{7}7\mu = \mu \end{aligned}$$

$$\begin{aligned} E(\hat{\Theta}_1) &= E\left(\frac{2X_1 - X_6 + X_4}{2}\right) = \frac{1}{2}(2E(X_1) - E(X_6) + E(X_4)) \\ &= \frac{1}{2}(2\mu - \mu + \mu) = \mu \end{aligned}$$

Both estimators are unbiased.

b

Which estimator is better, and in what sense is it better?

Hint: Calculate the MSE for each estimator.

✓ **Answer**

Since both estimators are unbiased, $MSE = Var$

$$\begin{aligned} Var(\hat{\Theta}_1) &= Var\left(\frac{X_1 + X_2 + \dots + X_7}{7}\right) = \frac{1}{7^2}(Var(X_1) + Var(X_2) + \dots + Var(X_7)) \\ &= \frac{1}{7^2}7\sigma^2 = \frac{\sigma^2}{7} \end{aligned}$$

$$\begin{aligned} Var(\hat{\Theta}_1) &= Var\left(\frac{2X_1 - X_6 + X_4}{2}\right) = \frac{1}{2^2}(2^2 Var(X_1) + Var(X_6) + Var(X_4)) \\ &= \frac{1}{2^2}(2^2\sigma^2 + \sigma^2 + \sigma^2) = \frac{3}{2}\sigma^2 \end{aligned}$$

Since $\frac{\sigma^2}{7} < \frac{3}{2}\sigma^2$, meaning its MSE is lower, $\hat{\Theta}_1$ is a better estimator.